

Number Theory

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This class by Shreyash Sharma was focused on Number Theory.

§1 Problem

Problem statement

Let $a_1 < a_2 < \dots < a_{2017}$ and $b_1 < b_2 < \dots < b_{2017}$ be positive integers such that

$$(2^{a_1} + 1)(2^{a_2} + 1) \dots (2^{a_{2017}} + 1) = (2^{b_1} + 1)(2^{b_2} + 1) \dots (2^{b_{2017}} + 1)$$

Prove that $a_i = b_i$ for $i = 1, 2, \dots, 2017$.

¶ **Solution.** We can expand the terms on each time to get.

$$\begin{aligned} & (2^{a_1} + 2^{a_2} + \dots + 2^{a_{2017}}) + (2^{a_1+a_2} + 2^{a_1+a_3} + \dots + 2^{a_{2016}+a_{2017}}) \\ & \quad + (2^{a_1+a_2+a_3} + \dots) + \dots \\ & \quad + 2^{a_1+a_2+\dots+a_{2017}} + 1 \\ & = \\ & (2^{b_1} + 2^{b_2} + \dots + 2^{b_{2017}}) + (2^{b_1+b_2} + 2^{b_1+b_3} + \dots + 2^{b_{2016}+b_{2017}}) \\ & \quad + (2^{b_1+b_2+b_3} + \dots) + \dots \\ & \quad + 2^{b_1+b_2+\dots+b_{2017}} + 1 \end{aligned}$$

Now, pulling out 2^{a_1} and 2^{b_1} left and right side respectively we get,

$$\begin{aligned} 2^{a_1}(\text{something odd}) &= 2^{b_1}(\text{something odd}) \\ \implies a_1 &= b_1 \end{aligned}$$

Now, we have reduced our problem to,

$$(2^{a_2} + 1) \dots (2^{a_{2017}} + 1) = (2^{b_2} + 1) \dots (2^{b_{2017}} + 1)$$

Thus, we can use this argument again to get $a_i = b_i$ for all $1 \leq i \leq 2017$.

Problem statement

The sequence (a_n) is defined by $a_1 = 1$ and $a_n = n(a_1 + a_2 + \cdots + a_{n-1})$, $\forall n > 1$.

- (a) Prove that for every even n , a_n is divisible by $n!$.
- (b) Find all odd numbers n for the which a_n is divisible by $n!$.

¶ Solution. Since $a_{n-1} = (n-1)(a_1 + \cdots + a_{n-2})$ we can replace it. Thus we get

$$\begin{aligned} a_n &= n(a_1 + \cdots + a_{n-2} + (n-1)(a_1 + \cdots + a_{n-2})) \\ &\implies a_n = n^2(a_1 + \cdots + a_{n-2}) \end{aligned}$$

Again, since $a_{n-2} = (n-2)(a_1 + \cdots + a_{n-3})$ we get

$$\begin{aligned} \implies a_n &= n^2(a_1 + \cdots + a_{n-3} + (n-2)(a_1 + \cdots + a_{n-3})) \\ &\implies a_n = n^2(n-1)(a_1 + \cdots + a_{n-3}) \end{aligned}$$

we can keep going and we will get a formula for a_n in terms of n ,

$$\begin{aligned} a_n &= n^2 \cdot (n-1) \cdot (n-2) \cdots 3 \\ &\implies a_n = \frac{n \cdot n!}{2} \end{aligned}$$

Thus, for even n we can see that $n!$ indeed does divide a_n .
And the only odd n for which $n!$ divides a_n is 1.